

Sangyu Han

Seoul, South Korea

acoexist96@snu.ac.kr ✉ Google Scholar 📄 +82 10 9429 0388 📞

Career Objective

I am deeply interested in **Explainable AI**, particularly in **Mechanistic Interpretability** (i.e., reverse-engineering neural networks). I aim to identify monosemantic representations within a model's hidden layers and uncover the **mechanism of the interlayer interaction**. My goal is to enable anyone to easily observe a model's internal representation processes, thereby contributing to the development of safe, transparent, and trustworthy AI.

Education

- **Seoul National University**, Seoul, Republic of Korea *Aug. 2022 – Present*
Combined MS/Ph.D of Interdisciplinary Program of Artificial Intelligence
- **Korea University**, Seoul, Republic of Korea *Mar. 2016 – Aug. 2022*
B.S. in Health and Environment Science
Interdisciplinary Major in Artificial Intelligence

Publications

1. **Decompose the model: Mechanistic Interpretability in Image Models with Generalized Integrated Gradients (GIG).**
Yearim Kim*, Sangyu Han*, Sangbum Han, Nojun Kwak.
Under review for ICLR 2025
2. **Bi-ICE: An Inner Interpretable Framework for Image Classification via Bi-directional Interactions between Concept and Input Embeddings**
Jinyung Hong*, Yearim Kim*, Keun Hee Park, Sangyu Han, Nojun Kwak, Theodore P Pavlic.
Under review for CVPR 2025, 2024
3. **Respect the model: Fine-grained and Robust Explanation with Sharing Ratio Decomposition**
Sangyu Han*, Yearim Kim*, Nojun Kwak.
ICLR, 2024
4. **Feature extraction on Convolutional Neural Network Layers with Channel-Wise Neuron Vector**
Sangyu Han, Gilsang Yoo.
International Conference on Innovation Convergence Technology, 2020

Teaching Experience

- **Samsung Electronics, Teaching Assistant** *Sep. 2024*
Conducted teaching assistance for "Explainable and Responsible AI".
- **Korea Data Agency, Teaching Assistant** *Jun. 2021 – Sep. 2021*
Assisted with the "Curriculum on Intelligent Information System Development" project.

Projects

- **Predicting academic performance of students by revealing socio-latent features, Korea University** *Dec. 2021 – Jun. 2022*
Developed a predictive model for academic performance using 6 years of poll data.
- **Sleep stage classification from non-EEG PSG data using long short-term memory neural networks, Korea University Medical Center** *Dec. 2021 – Jun. 2022*
Predicted sleep stages using non-EEG polysomnography data from MAXIM smart band.

- **Development of a stereo image-based environmental recognition algorithm for lunar exploration mobility, Hyundai Motor** *Jul. 2023 – Jul. 2024*
Generated bird-eye-view maps of lunar surfaces with stereo-based 3D object detection.

Awards

- **Dean's List, Korea University** *2nd Semester, 2020*
Honors for students with a cumulative GPA of at least 4.50/4.50

Skills

- **Languages:** Korean (Native), English (Advanced)
- **Technical Skills:** Python (Pytorch, Numpy), C++